superfloat.

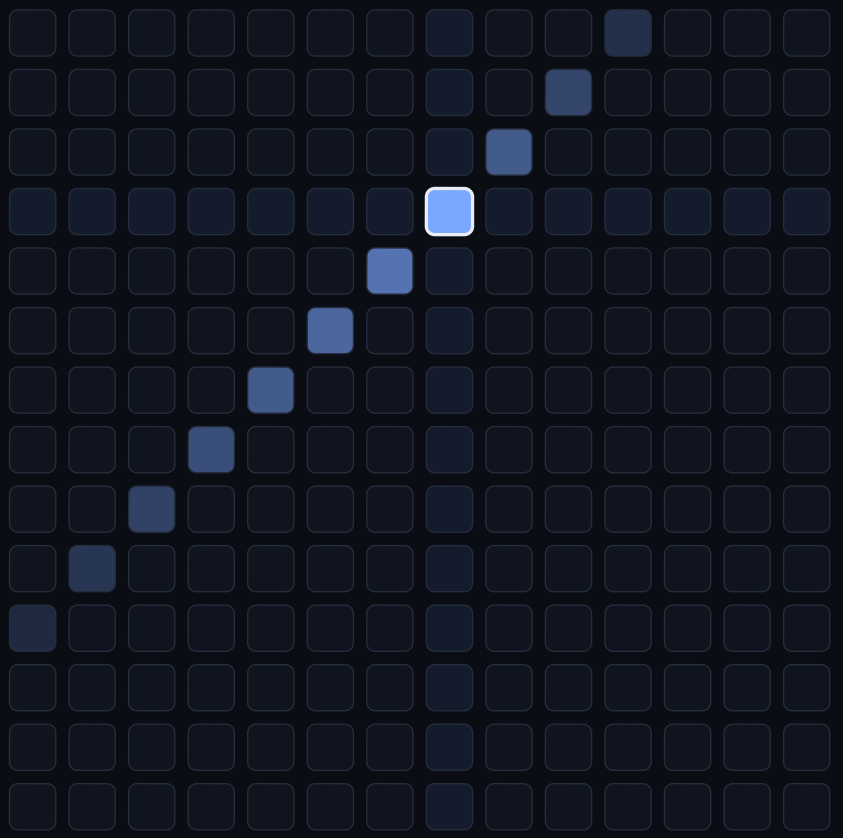
PARTNER BRIEFING · ATREIDES 2 NM

Atreides.

Ready for 2 nm.

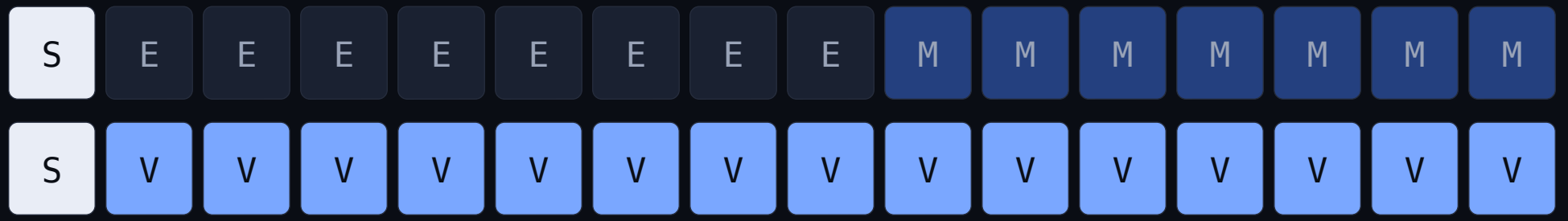
Where the SF16 accelerator stands, what the 2 nm program costs, and what we need from foundry and design-service partners.

SIGNAL FLOW / SF16



PE[3,7] · C ← C + A · B

Superfloat spends every bit on value



S sign · E exponent · M mantissa · V value, signed fixed point Q1.15

99.999%

of parameters in trained networks already sit inside [−1, 1]. The exponent pays for range they never use.

No exponent

hardware in the multiply-add: no align, no normalise, one pipeline stage on the partial-sum path.

SF4 to SF16

one format family at any width, converted exactly from standard floats. SF8 lands within 1% of SF16.

Three chips signed off on SkyWater 130 nm

GPT-2 small, same RTL	SF16	FP16	BF16
Tile area	1.40 mm²	1.86 mm²	1.63 mm²
Chip power, decode, 50 MHz	251 mW	460 mW	441 mW
Energy per output token	5.55 J	10.17 J	9.74 J
Energy per YOLOv8n image	53 J	98 J	94 J

SAME CYCLES, LESS ENERGY

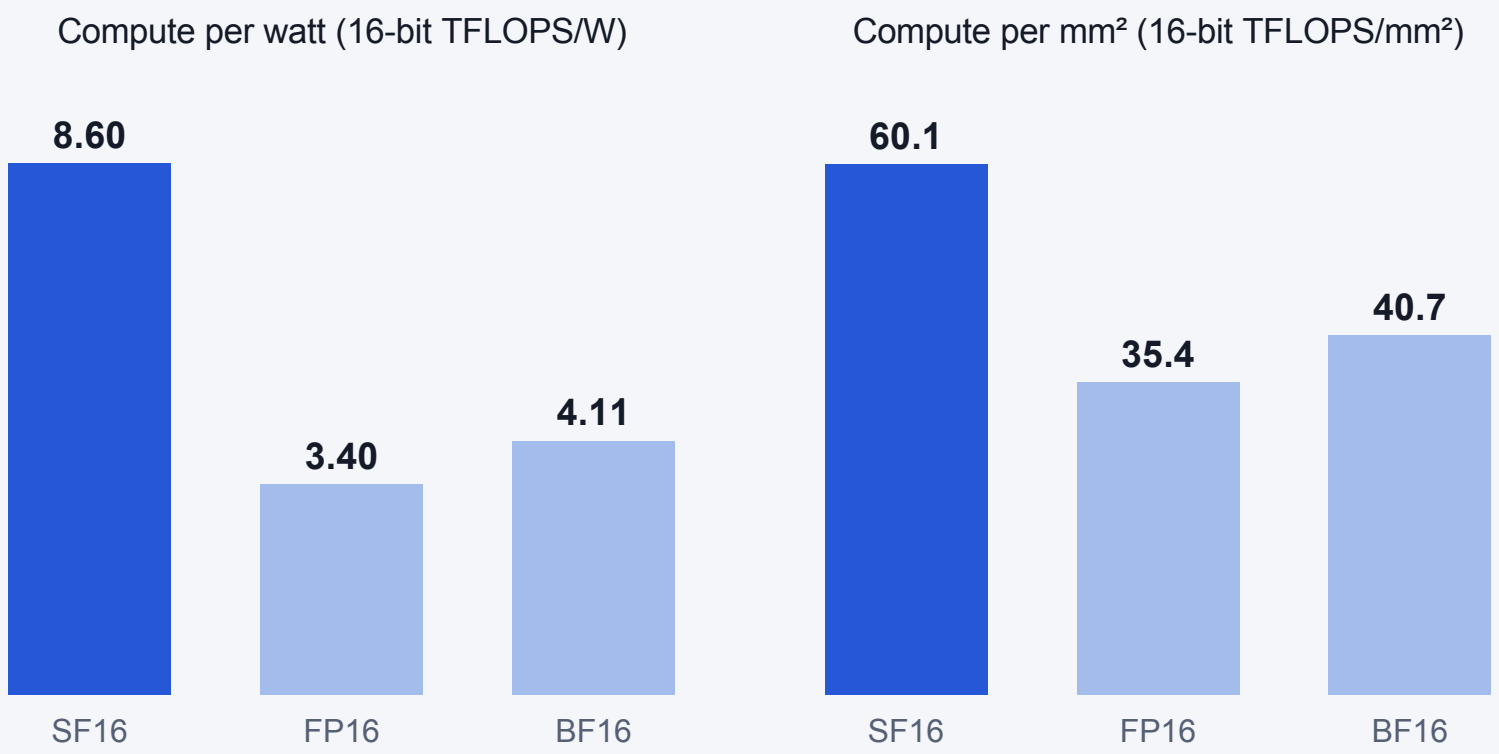
−45%

energy per token against FP16, and 43% against BF16, in prefill, decode and YOLOv8n alike.

Each chip: four tiles of 8×8 systolic arrays (256 MACs), a custom 16-bit ISA, SIMT threads. Zero violations in every sign-off check, timing met across nine corners. **Next:** the SF16 chip on ChipFoundry shuttle CI2612, tape-out 7 December 2026.

ChipFoundry OpenFrame v1, signed off 1 Oct 2026. Post-layout power and energy; only the PE, FMA and activation units differ.

2.5× the compute per watt of FP16



TOP CLOCK

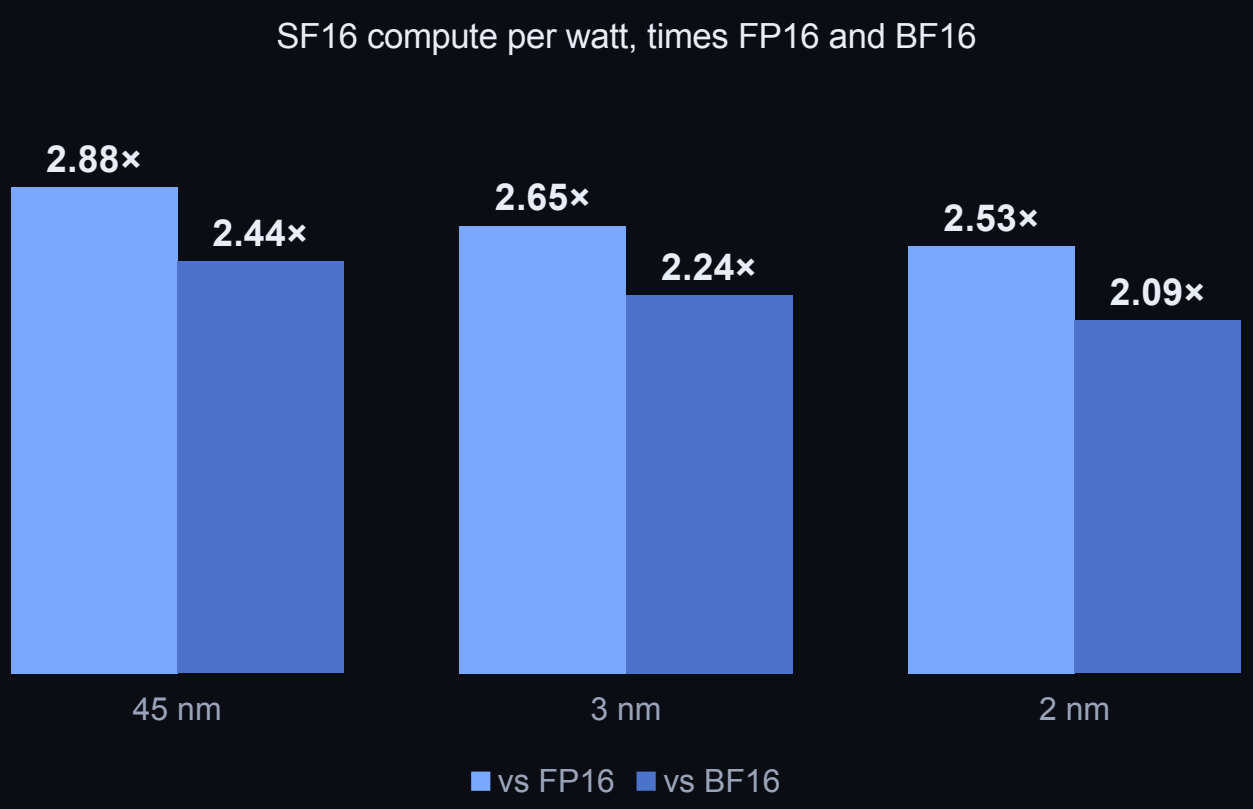
3.39 GHz

with the multiplier over two stages, against 3.01 GHz for FP16 and 3.03 GHz for BF16. SF16 has no float align or add stage to hold it back.

233 fJ per MAC against 589 (FP16)

v3 processing elements on the GT2N 2 nm predictive PDK (not a TSMC PDK): OpenROAD to global route, 0.7 V, typical corner, real GPT-2 operands at 2 GHz. Bit-exact at gate level.

The lead holds at 45, 3 and 2 nm



HAMMERBLADE MANYCORE

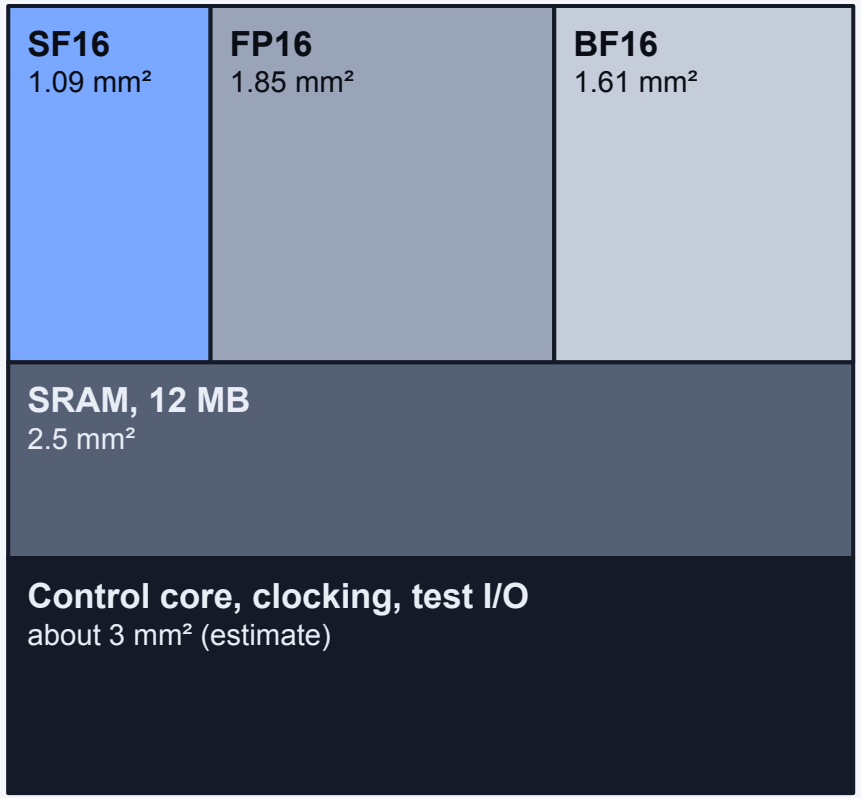
All 180 GPT-2 prefill matrix multiplies bit-exact on a RISC-V manycore, 11% sooner to the first token.

ACCURACY

GPT-2 perplexity within 0.55% of FP32. Detection within about 4% of FP32; classification and video level or better.

Same flow and GPT-2 stimulus at every node. NanGate45 and GT3 are research PDKs, one corner each, global-route parasitics.

A 10 mm² N2 shuttle chip comes first



Proves the SF16 matrix unit, SRAM and clock in TSMC silicon before the production mask set.

Compares all three formats on one die, one wafer: one 128×128 matrix unit each.

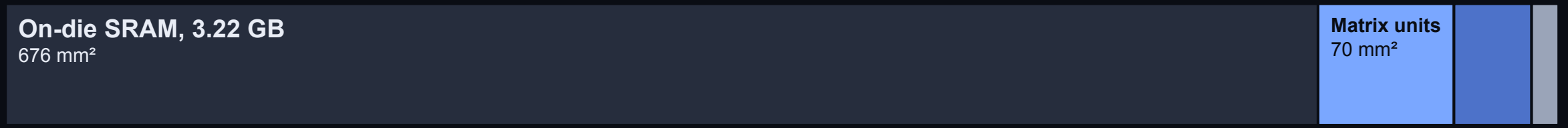
Yields dozens of dies from one shuttle; we plan around 40 for bring-up and validation.

Tapes out on the first 2027 N2 shuttle (March, if TSMC keeps its March and September runs).

Costs about $800K to $1.6M in silicon (estimate), about $21K to $40K a die.

Matrix-unit areas from the measured 2 nm blocks; SRAM at TSMC N2's published 38.1 Mb/mm². Floorplan drawn to scale.

800 mm², 1.73× the matrix compute of FP16



I/O and HBM PHYs 40 mm² · vector units, control, DMA, network 14 mm²

4,194

peak 16-bit TFLOPS from 64 matrix units, against 2,425 (FP16) and 2,818 (BF16) in the same 70 mm²

4.8 TB/s

from four HBM3E stacks, 144 GB, for what the 3.22 GB of SRAM cannot hold

1.66×

GPT-2 small decode at batch 1: 44,403 tokens/s against 26,743 for FP16

Before production tape-out: much wider vector units, faster weight loading for decode, and more capacity for long-context batches. Each applies to all three formats.

Modeled: 2 GHz, 700 W cap, roofline over matrix, vector, memory and power. Only the matrix units come from measured 2 nm blocks.

Test chip in 2027, production from 2028

NOV TO DEC 2026

Start

Design-service partner, TSMC NDA, N2 PDK; reserve the shuttle slot; send the 2028 wafer forecast.

7 DEC 2026

130 nm tape-out

The SF16 chip on ChipFoundry CI2612; parts back May 2027.

MARCH 2027

N2 test chip

Tape-out on the first 2027 N2 shuttle.

MID-2027

Bring-up

Dies back, measured against the 2 nm models. Second shuttle in September if a respin is needed.

Q4 2027

Production tape-out

The 800 mm² die on its own mask set.

2028

Year 1

1,000 N2 wafers, about 40,000 chips, first parts mid-year.

Why start now: N2 capacity is booked into 2028 and reserved about a year ahead with prepayment, so 2028's wafers must be requested before the test chip is back.

$29M to $70M to production tape-out

Phase	What it covers	Estimate
130 nm shuttle	The SF16 chip on ChipFoundry ($14,950), boards and bring-up	**$30K**
N2 test chip	10 mm² shuttle silicon, packaging and boards, design-service allowance, respin reserve	**$1.2M to $4.4M**
Production tape-out	N2 mask set, IP licences, design-service, test program and probe cards	**$23M to $59M**
EDA tools	About 10 seats at $80K to $150K an engineer-year	**$800K to $1.5M**
Team	Six hires: three in VLSI, three in AI	**$3.8M to $4.6M**
Total		**$29M to $70M**

WHERE IT GOES

~70%

is the N2 mask set ($15M to $30M) and IP licences ($5M to $20M).

Design-service NRE has no public figure; it is an allowance until a partner quotes it.

N2 figures are estimates from public reports; TSMC publishes no prices. Excludes year-1 wafers and capacity prepayment.

About $3,550 per chip in year 1

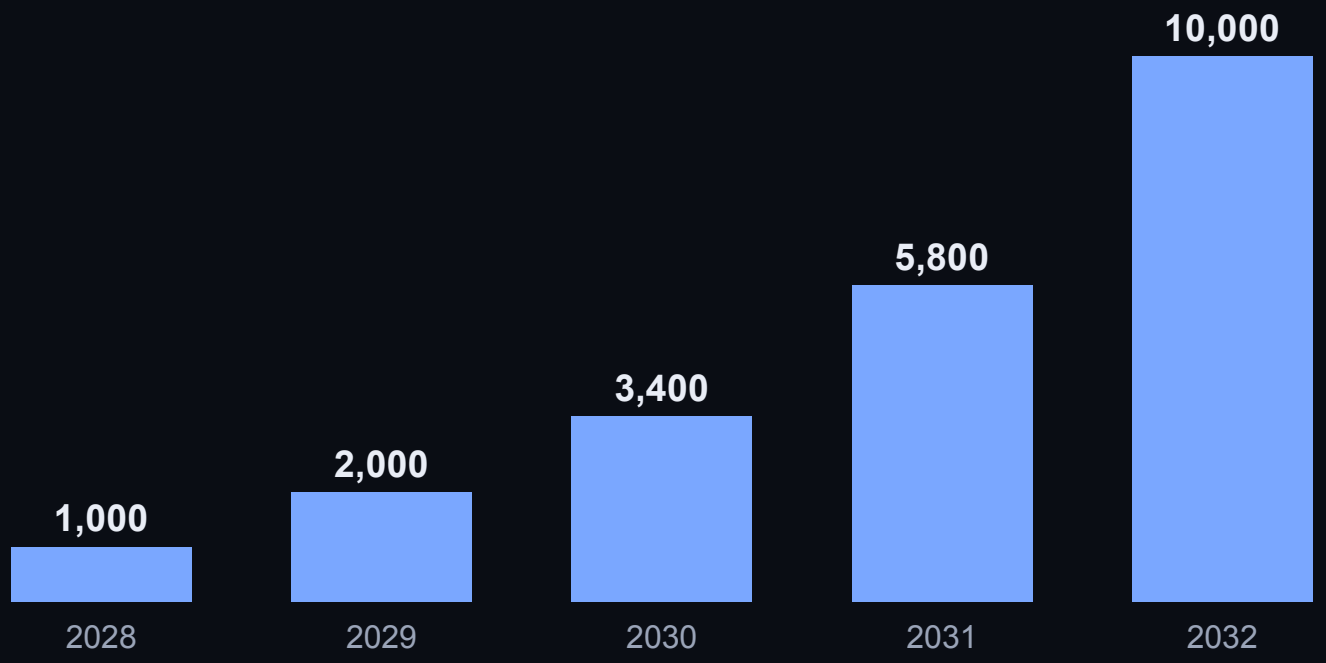


Per chip	Range
Silicon	$500 to $1,000
HBM3E, four 36 GB stacks	$920 to $1,840
CoWoS-class packaging	$600 to $1,200
Assembly and test	$370 to $600
Per chip, at the 40-die plan	**$2,640 to $4,390**

Year 1, 40,000 chips: $142M at plan ($106M to $176M), of which $30M is 1,000 N2 wafers.

Per chip at 40 good dies a wafer. HBM3E and CoWoS from public estimates; contract prices are confidential and rising.

1,000 wafers in 2028, 10× by 2032



Year	Chips	Wafers
2028	**40,000**	$30M
2029	**80,000**	$60M
2030	**136,000**	$102M
2031	**232,000**	$174M
2032	**400,000**	$300M

Plan targets, not orders. A 100 mm² edge die would give about 550 chips a wafer.

Chips at 40 good 800 mm² dies a wafer (30 to 60 across the yield range). Wafers at today's $30,000; TSMC raised 2026 prices 5 to 10%.

Partners for access, sign-off, IP, packaging and test

Foundry access

TSMC NDA and N2 PDK, the 2027 shuttle slot, and 2028 wafer allocation.

N2 sign-off

N2-certified EDA, DFM and timing, power and reliability sign-off.

Third-party IP

HBM3E PHY and controller, high-speed I/O, PLLs and clocking.

Packaging

Flip-chip for the test chip; a CoWoS-class interposer with four HBM3E stacks for production.

Test

Wafer sort, final test and probe cards, for dozens of dies and then 40,000 a year.

Quotes

Shuttle slot, mask set and 2028 wafers, to replace our public-report estimates.

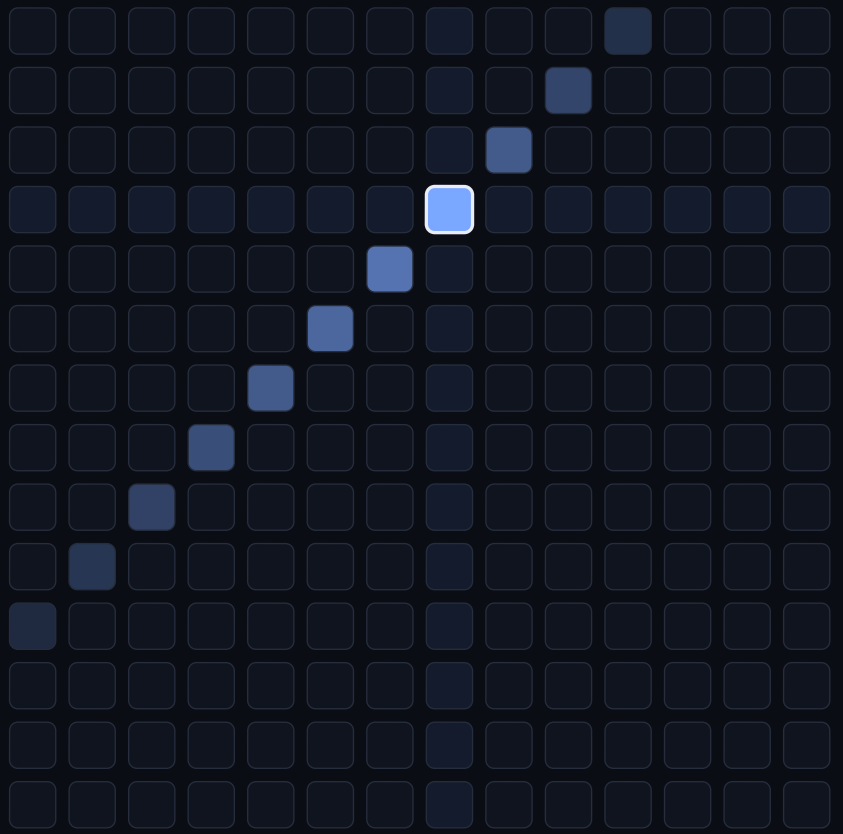
We own **architecture, RTL, verification and physical design.** At first-year volume, a Taiwan design-service partner close to TSMC is the practical route; direct engagement as volumes grow.

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October 2026